DISK DEMON

{ Based on original Datel Disk Demon documentation }

Datel Disk Demon – the fastest parallel disk operating system for Commodore 64 and 1541 disk drive

The Datel Disk Demon is an exciting product for the 1541 owner!!. If you own a 1541 you will know that although they are reliable in operation, they are very siow.

Fast load cartridges such as Diskmate and the disk turbo on Action Replay can speed up the process considerably with load times down to as low as 10 seconds for a typical game program. But this is about as fast as you can get with the 1541 standard system because it uses serial data transfer. This means that all the data is transmitted down just two wires which limits the speed.

The new DISK DEMON is a true parallel operating system for the 1541 and brings loading and saving speeds that were never dreamed of!!

Just imagine a typical game of 202 blocks LOADING IN 2.5 SECONDS.

Disk Demon is easy to install, consisting of two main assemblies - one plugs inside the disk drive and the other inside the 64. Usually, no soldering is required. Now plug the parallel cable into the user port of the 64 and that's it. You have the fastest disk operating system available for the 1541.

Just look at the speeds:

* LOADS PROGRAM FILES AT 60 TIMES NORMAL SPEED (202 blocks, 2.5 secs)

* SAVES PROGRAM FILES AT 25 TIMES NORMAL SPEED (202 blocks, 8 secs)

* LOADS SEQ FILES AT 26 TIMES NORMAL SPEED

* SAVES SEQ FILES AT 20 TIMES NORMAL SPEED

Disk Demon is not only a superfast system, it also features extra disk commands to do all the things that should have been done originally... plus many more.

*** 16 FUNCTION KEYS.**
Including LIST ,SAVE, LOAD, RUN, DIRECTORY, OLD, VERIFY, FAST FORMAT, GOODBYE, etc.

*** FORMAT DISKS UP TO 749 BLOCKS.**

*** LOCK/ UNLOCK FILES.**

*** 'FLOAD' COMMAND**
LOADS FILES UP TO 250 BLOCKS.

NB. Other systems such as Dolphins DOS can't cope with files larger than 202 blocks. This makes Disk Demon super-powerful when used with Action Replay III. ie. An Action Replay backup will reload in UNDER 3 SECS!!

*** BUILT IN FILE COPIER.**
The **'FCOPY'** command will copy files UP TO 250 BLOCKS at the press of a function key. It is super fast - the complete process takes about 15 seconds including the save.

* **User Port through bus** is supplied free to allow other devices to be connected at the same time (with Dolphin you paid £14.99 for that facility alone).

*** A HOST OF OTHER NEW COMMANDS**
including Number conversion, Unstoppable reset , ON/OFF etc, etc. Plus, the other disk operations including Scratch and Verify are also much faster.

*** DISK DEMON is highly compatible with commercial software.**
On the odd occasion where there is a problem it is switched out at the flick of a switch. The system will then return to the original DOS.

*** When you compare parallel systems remember that Disk Demon not only has more features than the others but it also works at at least TWICE THE SPEED of any other currently available. So, don't sit waiting for that disk to load, **GET DISK DEMON NOW**!!!

In 1987, DISK DEMON sold for £64.99 post free. A Utility disk is also available with 15 second disk copier, superfast selective file copier and file splitter. This was originally sold for £7.99 or £4.99 when bought with Disk Demon.

Instructions for Datel Disk Demon

When using the original Commodore DOS Kernal mode, the command and operation structure of the Commodore 64 remains unchanged.

In Disk Demon mode disk LOAD, SAVE, & VERIFY are all at Turbo speed and additional commands and capabilities are available.

FUNCTION KEYS

The standard function keys are assigned as follows:

f1	**LIST**	- Lists the Basic program currently in memory
f2	**OLD**	- Recover a BASIC program that has been deleted by 'New'
f3	**RUN**	
f4	**LIST OFF/ON**	
f5	**LOAD,8,1-** Fast Load the program next to the current cursor position	
f6	**SAVE"**	
f7	**@$**	- Display directory without overwriting program in memory)
f8	**LOAD, 8, 0**	- Fast Load first program into Basic memory start address
	See https://www.c64-wiki.com/wiki/LOAD for more info	

ADDITIONAL FUNCTION KEYS

F10 (CTRL-f1**)**	**Format** (@N*)	- Fast Format of disk in drive
	****** NOTE NO CONFIRMATION IS REQUESTED ******	
F11 (CBM-f3)	**FLOAD**	- see page 4
F12 (CTRL-f3)	**FCOPY**	- see page 4
F13 (CBM-f5)	**LOAD ,8,2**	- Load first program at reduced speed
F14 (CTRL-f5)	**VERIFY**	
F15 (CBM-f7)	**HELLO**	- enables DOS wedge
F16 (CTRL-f7)	**GOODBYE**	- exits DOS wedge

F16 (CTRL & f7 together) is **GOODBYE**.
This will disable BASIC enhancements but leave the fast disk access intact.
Restart with SYS 58529 or CBM-f7 (**HELLO**)

f4 lists BASIC programs where list protection has been implemented.
Repeated pressing of f4 will toggle this feature on/off.

See '**NEW FEATURES'** for more on Loading and Saving

NEW FEATURES

FCOPY & FLOAD commands

FCOPY will copy any file up to 250 blocks.
Simply list the directory **(f7)** and move the cursor up to the file that you wish copy.
Type '**FCOPY'** (or use **CTRL-f3**) at the beginning of the line and then press RETURN
The file will be loaded then you will be prompted to insert another disk to which the file is to be copied.

FLOAD allows the user to load files over 202 blocks.
Simply list the directory **(f7)** and move the cursor up to the file that you wish copy.
Type '**FLOAD'** (or use **CBM -f3**) at the beginning of the line and press RETURN
The file will be loaded.

Number Conversion

Hex, Binary and Octal Numbers can now be converted to decimal by prefixing the number with **'$**', **'%**', or **'&**'

HEX to DECIMAL	eg PRINT $1000 = 4096
BINARY to DECIMAL	eg PRINT %1000 = 8
OCTAL to DECIMAL	eg PRINT &1000 = 512

Loading and Saving

Disk Load and Save commands have been enhanced and the following additional options are now available. Note that files saved with the Disk Demon are saved with a sector interleave of 1

All of the following commands should be ended by pressing the **RETURN** key

1) LOAD
Use Disk Demon to load the first file on disk and don't relocate it (same as Load "*",8,1)

2) **LOAD " XYZ"**
Use Disk Demon to load file "**XYZ**" with no relocation (same as Load "XYZ",8,1)

3) **LOAD "XYZ",8**
Use Disk Demon to load file " **XYZ**" and relocate it to the start of BASIC ($801 - same as Load "X",8)

4) **LOAD "XYZ",8,2**
Same as 2, but don't use Disk Demon, just use parallel port.

5) **LOAD "XYZ",8,3**
Same as 4 but ALL subsequent loads do not use the Disk Demon (same as sending '@-' before loading)
NB This does not appear to work with the original Disk Demon kernel ROM

6) **LOAD "XYZ",8,4**
Same as 2 but 'end of BASIC' pointers are not changed.

7) **LOAD "XYZ",8,0,xxxxx**
Load file **XYZ** to memory address **xxxxx**
xxxxx can be in decimal or hex. Prefix hex numbers with "$"

8) **SAVE "XYZ",8,0,xxxxx,yyyyy**
Save memory to a file called **'XYZ'**
'**xxxxx'** is the start address of the memory to save and '**yyyyy'** is the end address plus 1.
(**xxxxx** & **yyyyy** can be in decimal or hex. Prefix hex numbers with "$")

Any access of a non-parallel drive will default to the old slow serial routines.

Disk Demon Commands

@ Show status of disk drive
@ {*command*} - executes the ***{command}*** and shows status
eg **'@I'** will initialise the disk the in drive

On first starting the Commodore 64 and disk drive, '**@**' should report

73, PROFESSIONAL 41L2, 00, 00

@ D? - Shows current operating mode of Disk Demon in 1541 drive

The default mode at startup is **1Y+3S+L-**

Operating mode status descriptions

1 *1MHz & 2MHz processor speed (default)* / **2** *(2MHz only processor speed)*
Y *Verify when saving (default)* / **N** *(Do not Verify when saving)*
+ *Extra RAM Buffer Enabled (default)* / **-** *(Extra RAM Buffer Disabled)*
3 *Format 35 Tracks (default - see note)* / **4** *(Format 40 Tracks - see note)*
S+ *Fast Save Enabled (default)* / **S-** *(Fast Save Disabled)*
L- *Load with screen enabled (default)* / **L+** *(Faster Load with screen disabled)*

Disk Demon settings options

@D- Disable drive acceleration

***** WARNING *** DO NOT SAVE A FILE WHILE DRIVE ACCELERATION IS DISABLED, IT WILL CORRUPT THE FLOPPY DISK. ALWAYS ENABLE DRIVE ACCELERATION BEFORE SAVING. LOADING works normally, it is only saving that has problems**

@D+ Enable drive acceleration

@D1 Set processor speed to 1MHz with variable clock frequency
@D2 Lock processor speed at 2MHz

@D3 Format disks with 35 tracks.
This does NOT do a format. It sets the number of tracks to **35** when a format is performed
Disks formatted to 35 tracks will have a '2A' identifier.

@D4 Format disks with 40 tracks.
This does NOT do a format. It sets the number of tracks to **40** when a format is performed
Disks formatted to 40 tracks will have a '4A' identifier.

@DN Disable Verify when saving
@DY Enable Verify when saving

@DS- Turn off Fast Save
@DS+ Enable Fast Save

@DL+ Ultra Fastload with screen switched **off**
Note - this does not work correctly with the original Disk Demon C64 Kernal ROM but does work with the Professional DOS C64 ROM (included in'ROMs' folder). The drive becomes erratic after a @DL+ command has been used with the original Disk Demon C64 ROM

@DL- Normal Fastload with screen switched **on**

@8 or **@9** sets current drive number

@ D0 - Disable Disk Demon on 1541 disk drive and return to Commodore 1541 DOS 2.6

NB The drive must be reset to restart Disk Demon after @D0 command

Additional Disk operations

@DE	Expand a 35 track disk to 40 tracks without losing existing data. Also changes disk identifier from '2A' to '4A'
@D:{name}	Protect/Unprotect file {name} from deletion
@DH:{name},{ID}	Changes the disk name and header without erasing the contents
@DI	Changes automatic id on 40 track disks formatted on compatible systems

**********CHECK

@D: {filename}	Lock {filename} and prevent the file from being deleted. Repeat to unlock
@D: *	Locks all files and prevents them from being deleted. Repeat to unlock

Using Verify will double as a program locator because it prints the start and end address of the program.

All operations will work with both drive 8 or 9 (provided Disk Demon Board is installed)

To resume BASIC enhancements either Reset or type **SYS 58529**.

Unstoppable reset routine

If you have a reset button, press and hold SHIFT key while pressing the reset button.
It will even by pass any resident cartridges.

COMPATIBILITY

Disk Demon will speed up LOAD/SAVE/VERIFY significantly depending on program type.
However it should be noted that many disk games already have a fast loader and Disk Demon may not be able to speed them any further.

Normally in these cases there will be no clash and the disk will load as normal. On some occasions there may be a conflict which will mean that you may have to return to the standard Commodore 64 kernal and/or disable the Disk Demon expansion board completely with the command **'@D0'**

NB The drive must be reset to restart Disk Demon after @D0

*** IMPORTANT NOTE ***

The space for the extra functionality has been made by removing ALL tape and some RS232 operations so these are not available in 'Disk Demon' mode.

If you want to load from tape then switch back to the standard Commodore 64 kernal.

INSTALLATION - Commodore 64

***** TURN OFF ALL DEVICES *****

Remove the Kernal ROM from the **U4** socket on the Commodore 64

Install the Disk Demon Kernal ROM into the **U4** socket of the Commodore 64 using the EPROM adapter.

Disconnect all connections to/from the floppy drive and turn on the computer.

Within a second, the computer should display a solid blue screen with the following text

DATEL ELECTRONICS
PROFESSIONAL-DOS7
38911 BASIC BYTES

READY

The first two lines of text should be in light blue
The second two lines of text should be in white

A white cursor should be flashing underneath the letter 'R' in the word 'READY'.
If not, turn off and check the installation of the Kernal ROM

If the normal Commodore 64 start-up screen is displayed then flick the switch on the wire attached to the kernal ROM and restart the Commodore 64.

INSTALLATION – 1541 Disk Drive

***** TURN OFF ALL DEVICES *****

Disk Demon Board Installation

Remove the following chips from your 1541 drive:

UB4 (901229)
UC4 (6502)

If you do not remove the UB4 chip, the Disk Demon will not work !

Insert the **6502** processor (from **UC4**) into the long socket on the Disk Demon board with the notch facing towards the back of the drive.

Plug the Disk Demon board into the empty socket (**UC4**) on the 1541 System board (ie the socket from which the **6502** processor was removed).

NOTE The UB4 socket is left empty so keep the chip removed from the UB4 socket somewhere safe

Testing the Disk Demon Board installation

Switch on the 1541 drive.
The green light should come on and stay on.
The red light should come on for less than a second and then go out.

If the red light stays on for more than two seconds, switch off the drive and check that both chips are correctly inserted in their sockets and that the **UB4** (901229) chip has been removed.

Parallel Bus installation

Remove the **6522** chip from the socket marked **UC3**

Insert the **6522** chip from **UC3** into the 40-pin socket of the ribbon cable connector. Always pay attention to the direction of the chip notch which should be facing the ribbon cable.

Insert the 40-pin ribbon connector (with the **6522** installed) into the **UC3** socket with the ribbon cable facing towards the back of the drive

Testing the complete installation

***** TURN OFF ALL DEVICES *****

Attach the ribbon cable connector from the 1541 drive to the User Port on the Commodore 64
Attach a regular serial cable between the Commodore 64 and the 1541 drive

****** Note both the serial cable and ribbon cable must be connected ******

Turn on the 1541 drive

Turn on the Commodore 64

At the READY prompt, type **@** and press the '**RETURN**' key

The display should show

"73,PROFESSIONAL 41L2,00,00"

Next type **@D?** and press the '**RETURN**' key
The display should show

"0,1Y+3S+L-,0"

This means Disk Demon is installed correctly

Common Installation Problems

Red data light stays on when drive is switched on
- ROM chip **UB4** (901229) has not been removed
- Check the Disk Demon board is correctly inserted in the 1541 socket (UC4)
- Check the CPU is seated correctly in the Disk Demon board socket.
  Try removing both the board and 6502, applying contact cleaner and replacing
  This is the only common problem experienced during testing & development of the new board.
  If possible replace the original Commodore 'single wipe' socket on 1541 system board with either a 'double wipe' or machine pin socket.

@ Command responds with '73,C'
- ROM chip **UB4** (901229) has not been removed

No drive activity after pressing f7
- Make sure the User Port connector is correctly attached to the C64
- Check the Parallel Port adaptor board is correctly inserted in the 1541 socket (**UC3**)
- Check the **6522** chip is seated correctly in the Parallel Port adaptor board socket.
Try removing it, applying contact cleaner and replacing

Known issues

@D-
The original Disk Demon documentation states that the **@D-** command will disable drive acceleration. This seems to be partially true as load times increase by approximately 10%. However, trying to save a file after sending the @D- command will result in a corrupt file and may even corrupt the whole diskette.

@DL+
Use of the @DL+- command should disable the screen during program loading to allow for faster disk loads. With the Disk Demon Kernal ROM installed in the C64, this does not work. The screen is not disabled and the load will fail.

It does work correctly with the Professional DOS Kernal ROM installed in the C64

Multiple devices
Loading of some programs can be unreliable if an SD2IEC is also attached to the serial bus while Disk Demon is being used.